

FORM TP 2008192



TEST CODE **22220020**

MAY/JUNE 2008

CARIBBEAN EXAMINATIONS COUNCIL

ADVANCED PROFICIENCY EXAMINATION

ENVIRONMENTAL SCIENCE

UNIT 2: AGRICULTURE, ENERGY AND ENVIRONMENTAL POLLUTION

PAPER 02

2 hours 30 minutes

02 JULY 2008 (a.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. **DO NOT** open this examination paper until instructed to do so.
2. This paper consists of **NINE** questions, **THREE** from each Module.
3. Answer **SIX** questions, **TWO** from **EACH** Module.
4. Write your answers in the answer booklet provided.
5. The use of a silent, non-programmable, scientific calculator is permitted.

MODULE 1

Answer ANY TWO questions.

1. Figure 1 shows the agricultural yield from a farm when using inorganic fertilizers and organic fertilizers.

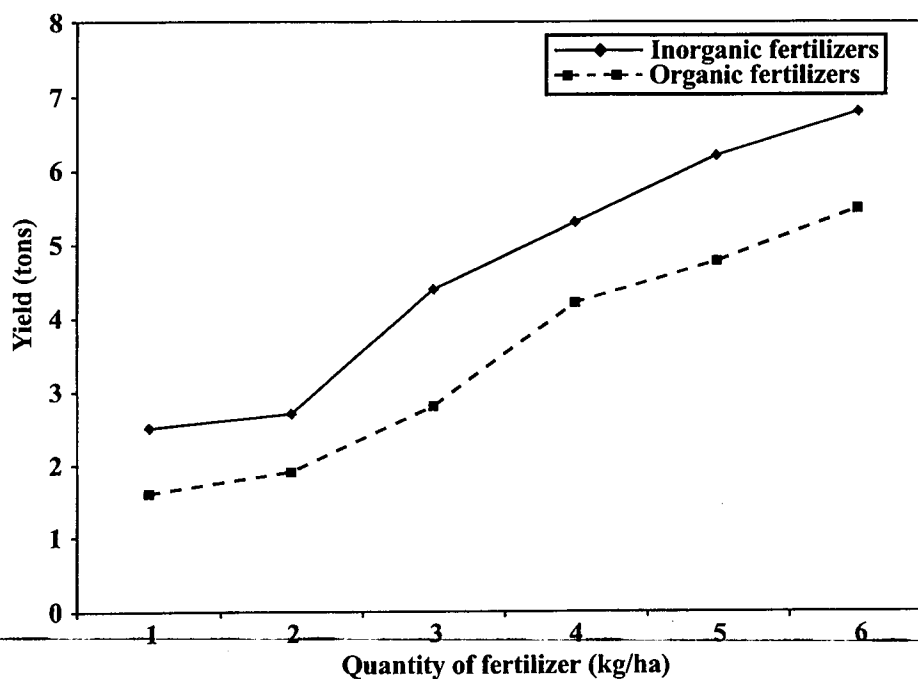


Figure 1. Yield and type of fertilizer used

- (a) With reference to Figure 1, state FOUR deductions that can be made about the agricultural yield from the farm. [4 marks]
- (b) Distinguish between 'organic' and 'inorganic' fertilizers. [4 marks]
- (c) Discuss why, in spite of the trend shown in Figure 1, farmers are still being encouraged to increase their use of organic fertilizers over inorganic fertilizers.

Include SIX points in your response.

[12 marks]

Total 20 marks

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2. Figure 2 shows the contributions made by subsistence and commercial agriculture to the gross domestic product (GDP), employment and foreign exchange earnings in Country A.

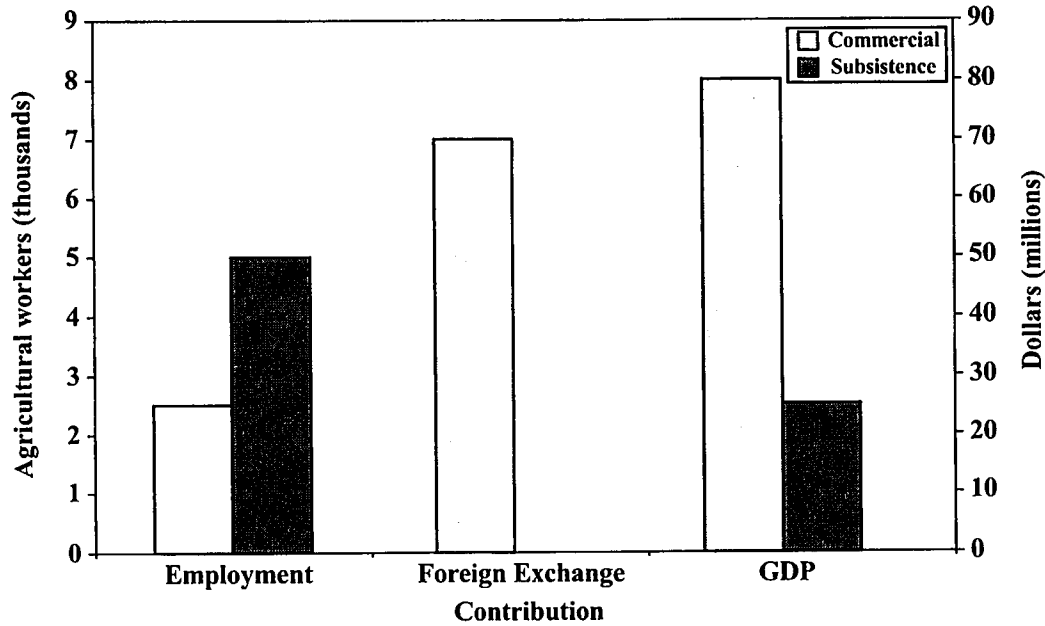


Figure 2. Contribution of agriculture in Country A

- (a) With reference to Figure 2, comment on the contributions of subsistence and commercial agriculture to employment, foreign exchange earnings and GDP in Country A. [4 marks]
- (b) How does mechanization account for the differences in the contributions shown in Figure 2? [12 marks]
- (c) Describe ONE feature of sustainable agriculture. [4 marks]

Total 20 marks

3. Hillside farming has been a common occurrence for years in Toco Village. Recently residents of the village have observed increased cases of soil erosion and degradation of the water quality. Farmers have been planting the same crops for years and have complained of decreased yields and increased pest infestation.
- (a) Residents in Toco Village were told by the agricultural extension officer that their farming practices were responsible for the increased cases of soil erosion and water quality degradation. Explain, using SIX points, why the officer would have said this to residents. [6 marks]
 - (b) Farmers in Toco Village were advised to practise crop rotation in an effort to improve yields and reduce the problems caused by pest infestation. Justify, using SIX points, the advice given. [6 marks]
 - (c)
 - (i) Name ONE soil conservation method that is appropriate for the farmers in Toco Village. [1 mark]
 - (ii) BRIEFLY describe the NAMED method in (c) (i) above AND the role it plays in soil conservation. [3 marks]
 - (d) Outline
 - (i) TWO features of no-tillage farming [2 marks]
 - (ii) why no-tillage farming is effective in reducing soil erosion. [2 marks]

Total 20 marks

MODULE 2

Answer ANY TWO questions.

4. (a) (i) List the types of fossil fuels. [1 mark]
- (ii) Outline the formation of fossil fuels. [3 marks]
- (b) Figure 3 shows the annual consumption of fossil fuels by developing countries in 1990, 2000 and 2003.

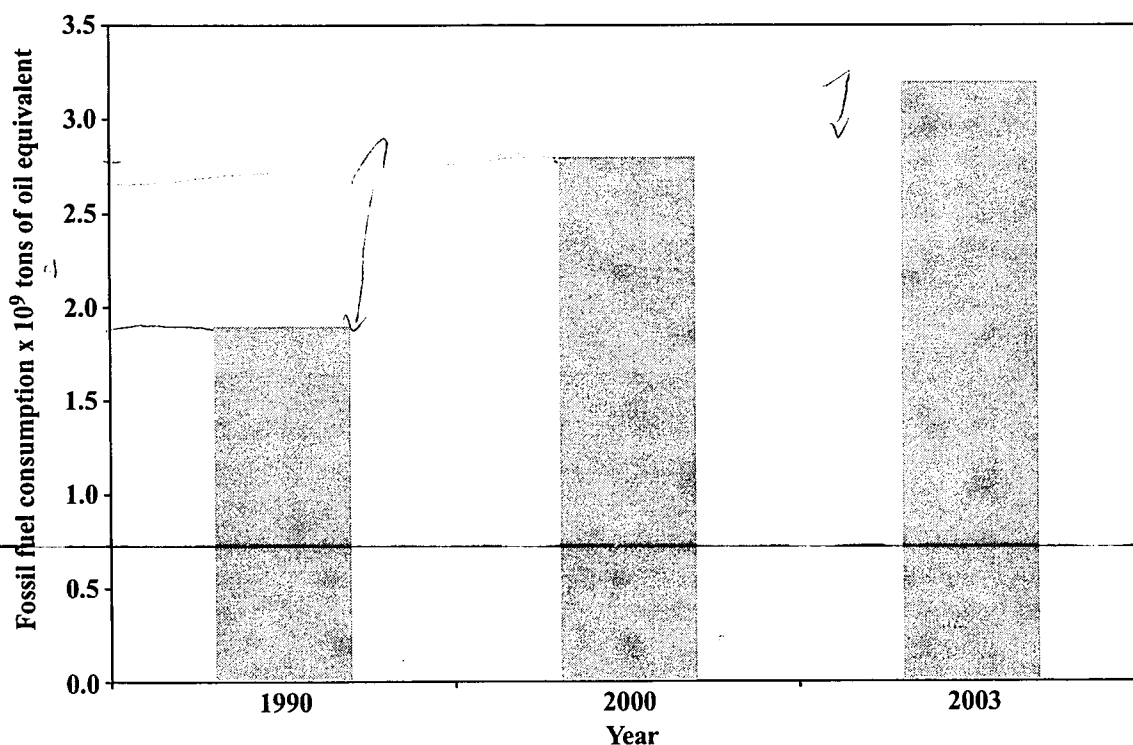


Figure 3. Annual consumption of fossil fuels by developing countries

- (b) (i) Compare the annual consumption of fossil fuel in 1990 with consumption in 2000 AND consumption in 2000 with that in 2003. [4 marks]
- (ii) Describe ONE environmental and ONE social impact on developing countries as a result of the consumption pattern indicated in Figure 3. [6 marks]
- (c) (i) Using an appropriate example, outline the meaning of 'demand management' in relation to energy use. [2 marks]
- (ii) Explain how 'demand management' can help mitigate the impacts of fossil fuel consumption. [4 marks]

Total 20 marks

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5. Study Figure 4 which illustrates a hydroelectric power plant and answer the questions that follow.

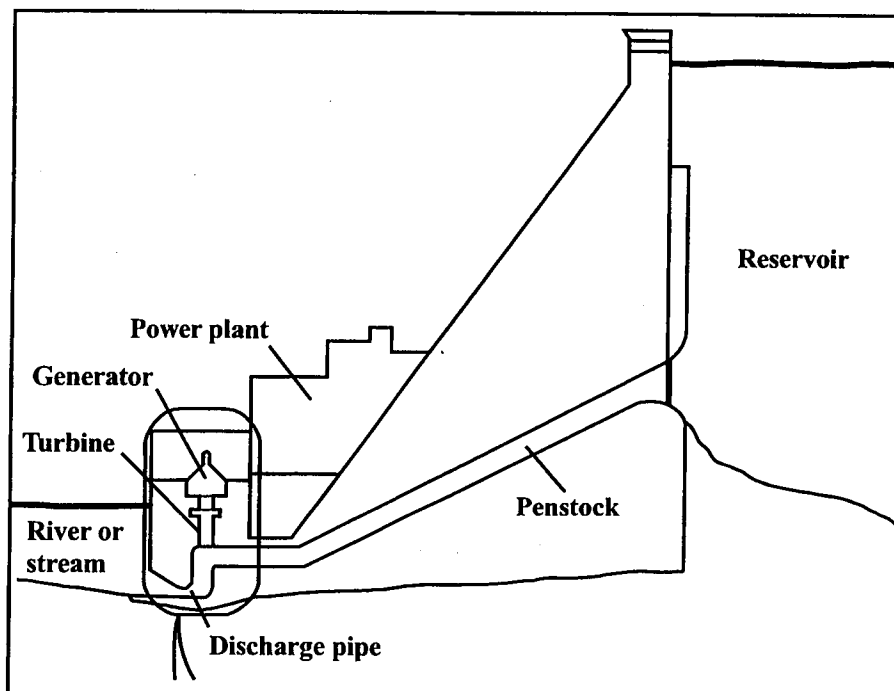


Figure 4. Hydroelectric power plant

- (a) Describe the energy conversion process occurring in the hydroelectric power plant, making clear in your description the meaning of the terms 'potential energy' and 'kinetic energy'. [4 marks]
- (b) State TWO advantages and TWO disadvantages of hydroelectric power generation. [4 marks]
- (c) Assess the suitability of (i) hydroelectricity AND (ii) solar energy to meet adequately the energy needs of developing countries.

Include SIX points in EACH response.

[12 marks]

Total 20 marks

6. (a) An incandescent bulb has a power rating of 60 W and an efficiency of 5%.
What is meant by
- (i) power rating of 60 W? [2 marks]
 - (ii) efficiency of 5%? [2 marks]
- (b) Outline TWO advantages of energy conservation. [4 marks]
- (c) Cogeneration may be utilised to reduce energy wastage.
- (i) What is meant by 'cogeneration'? [1 mark]
 - (ii) Outline how cogeneration is achieved. [2 marks]
 - (iii) Comment on the appropriateness of this system for use in Caribbean countries. [3 marks]
- (d) Explain how improving energy efficiency mitigates (reduces) environmental impacts resulting from energy use. [6 marks]

Total 20 marks

MODULE 3

Answer ANY TWO questions.

7. Table 1 presents results from a monitoring programme to determine levels of the pesticide dieldrin present in an aquatic ecosystem. The pesticide concentration was measured in each organism that formed a part of a complete food chain.

TABLE 1: CONCENTRATION OF DIELDRIN IN AQUATIC ECOSYSTEM

Category of Organism	Organism	Dieldrin Concentration (ppm)
Producers	Algae, aquatic plant	0.05
Primary consumers	Small fish	0.3 - 1.2
Secondary consumers	Large fish	1.4 - 2.6
Tertiary consumers	Large bird	4.0 - 6.0

Concentration of dieldrin in water = 0.0002 ppm

- (a) State FOUR inferences that may be drawn from the data presented in Table 1. [4 marks]
- (b) Outline TWO environmental pathways of the pesticide. [4 marks]
- (c) ~~Account for the difference in dieldrin concentration in the organisms presented in~~ Table 1. Include THREE points in your response. [6 marks]
- (d) Describe TWO characteristics of pesticides that account for their environmental impact. [6 marks]

Total 20 marks

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8. Figure 5 shows the solid waste composition for a Caribbean country in 1998 and 2007.

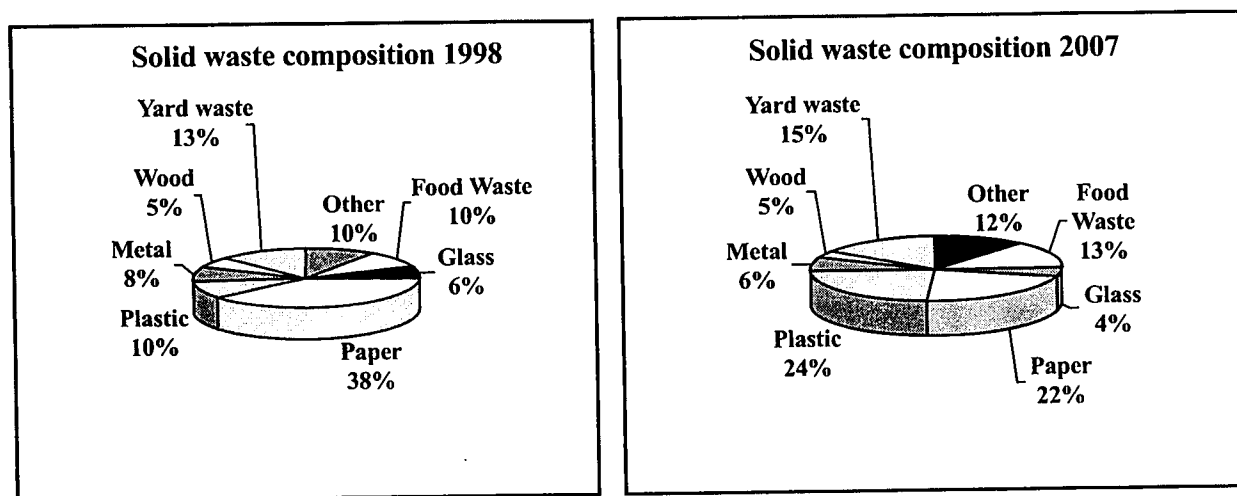


Figure 5. Solid waste composition for a Caribbean country, 1998 and 2007

- (a) State
- (i) TWO major changes in the composition of solid waste between 1998 and 2007
[2 marks]
 - (ii) the environmental significance of ONE of these changes in (i) above.
[2 marks]
- (b) Suggest TWO reasons for the change in the composition of solid waste between 1998 and 2007.
[4 marks]
- (c) Explain why it is necessary to minimise the amount of solid waste produced in Caribbean countries.
[4 marks]
- (d) Caribbean countries are proposing recycling programmes for minimising the volume of solid waste which they produce.
- (i) What is meant by 'recycling'? [2 marks]
 - (ii) Assess the effectiveness of their proposed programmes.
Include TWO advantages and TWO disadvantages in your response.
[6 marks]

Total 20 marks

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9. Figure 6 shows the variation in biological oxygen demand (BOD) and dissolved oxygen (DO) concentration with distance from a reference point established to monitor the water quality of a river.

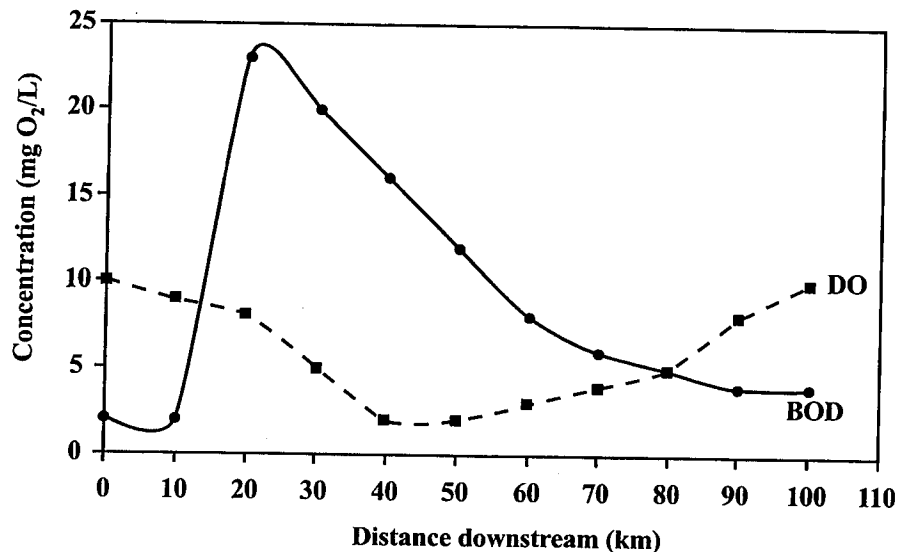


Figure 6. Variation in BOD and DO concentration

- (a) What information is provided by the BOD of a water sample? [3 marks]
- (b) Outline the laboratory process for determining BOD. [5 marks]
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- (c) Identify the type of water pollution source impacting on the river. Justify your response. [3 marks]
- (d) Account for the changes in DO concentration along the river. [5 marks]
- (e) At what distance from the reference point is it **LIKELY** that dead fish will be found floating in the river? Justify your answer. [4 marks]

Total 20 marks

END OF TEST